

Handling ensemble lists in Qt-DAB updated

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1 Introduction

Qt-DAB - with normal use - shows on its left part a long list of services, services from all ensembles seen. It resembles my car radio, we drove a number times from the Netherlands to Spain (and back) and my car radio shows dozens of service names from Belgium, France and Spain, services that will not run very well where I live.

It shows that some form of management of ensemble lists is needed. There are two forms, one is managing the current list, the other one is maintaining more than one list.

As said, adding to the lists is beyond user control: the software adds (channel service name) pairs whenever it detects one, so lists will grow.

In Qt-DAB, with its scanning possibilities one may need different service lists for different occasions and Qt-DAB provides the possibility of using different service lists on different occasions.

2 Running with different modes

Running the default mode The default setting is that the selector *show only audio services* is set, When running with default settings with input from a *real* device, Qt-DAB just collects all names of audio (and EPG/SPI) services and maintains a list of these names between program invocations. (The EPG/SPI services remain invisible and, when the channel such a service is on, the EPG/SPI services runs invisible in the background).

On selecting a service in ensemble mode or favorite mode, Qt-DAB may have to switch to a different channel, it may take a few seconds before the data describing the ensemble in that channel is collected.

Favourites In the list of services one may set a service to become *favourite*, just by clicking on the *third column in the row where the service is found*. The service is then marked by a small "*" in the third column.

The favourites - I usually have somewhere between 5 and 10 services marked as favourite - can be shown as a short list by switching to mode *favorites*. Obviously, the marking as favourite is stored in the database. (Removing a service from the list of favourites is by clicking on the star in the third column in ensemble mode.).

Handling changes in ensembles It happens from time to time that the content of an ensemble changes, last september, our national NPO replaced a couple of services by a single new one. Now the software is such that it incorporates automatically services not yet seen before. *In ensemble mode, when switching to a new channel, the software will automatically remove services from the ensemble in that channel that no longer exist.* In favourite mode this is not (yet) the case, but services can be removed from the list manually as well. Just click with the mouse on the channel name in the row where the service name is depicted, and the service will be deleted from both the display list as from the database.

Running with file input Qt-DAB supports reading from a file. Of course, when running with input from a file, a number of options is meaningless: i.e. selecting a channel, setting a favourite are options with no meaning. In case the channel on which the transmission was recorded was not detected (xml files and ".wav" files generated on Qt-DAB contain the channel information), no channel is shown.

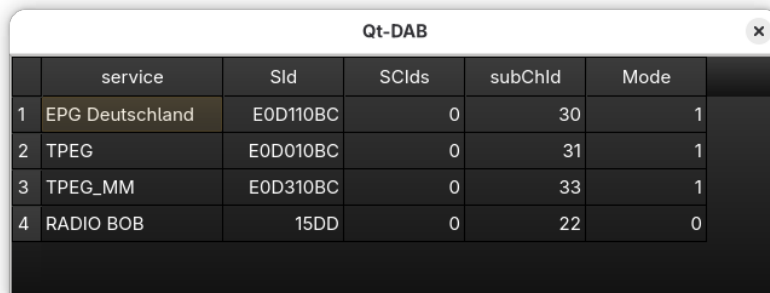
Running with packet services The *configuration and control* window contains a selector, labeled *show only audio services*, a selector by default *set*. If, however, the selector is *unset* there are two things different:

1. Only the services in the ensemble in the currently selected channel are shown, including:
2. packet services. such as TPEG and EPG are shown.

The rationale is that you most likely only want to see packet services is you want to select them. Note that EPG/SPI services are always selected. A selected packet service (usually) does not generate audio, most TPEG type services are encoded in a way Qt-DAB cannot decode them, but Qt-DAB extracts frames from the data and sends these frames as *udp* packets through a TCP port. The *port* through which the packets are transmitted is by default 8888, but can be set in the *configuration and control* window. Note that (sources for a) prototype program to extract the data from the *udp* packets is part of the source tree of Qt-DAB.

That is why in Qt-DAB the choice was made to run packet services (as it does with EPG/SPI services) as a background service, just letting it sending output to the aforementioned port. It allows to run several packet services simultaneously, think on a EPG/SPI service that runs in the background, and it allows an audio service to be selected as foreground task.

The obvious question then is when one or more services runs in the background, how to stop them. Of course the system will stop them when switching a channel, however, Qt-DAB provides also a simple controller, showing all running services, services running in background and foreground. *Touching the number, given as value for the number of services running (a hidden button on the Configuration and control window), shows a small separate window as shown below*



The image shows a window titled "Qt-DAB" with a close button (x) in the top right corner. Inside the window is a table with the following data:

	service	SId	SCIds	subChId	Mode
1	EPG Deutschland	E0D110BC	0	30	1
2	TPEG	E0D010BC	0	31	1
3	TPEG_MM	E0D310BC	0	33	1
4	RADIO BOB	15DD	0	22	0

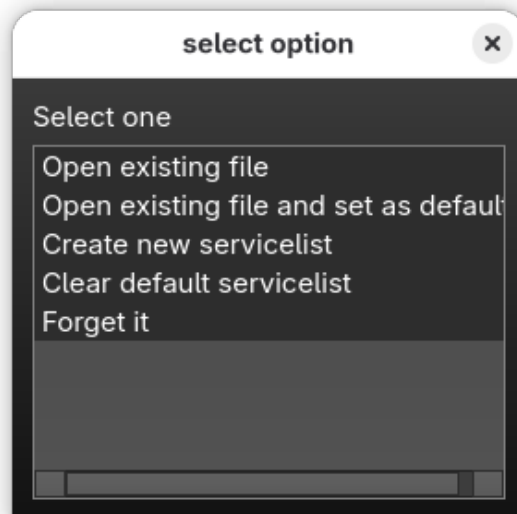
In this example, the table shows that 4 services are running. The foreground service is marked with Mode 0, the background services with Mode 1.

Clicking with the mouse on a background service, either automatically started or by the user, stops that service.

3 The ensemble list

As mentioned, ensemble lists are maintained between program invocations. The default list is stored and an xml encoded file in the directory *serviceLists*, stored in the Qt-DAB-files-7 directory in the home directory.

Setting the selector labeled *load selection* tells Qt-DAB that on the next (re)start you want to choose among a number of options for selecting a serviceList



The options are

1. *Open an existing file.* Apparently in a previous session a named servicelist was created and in the forthcoming session its data is used;

2. *Open an existing file and set as default.* Apparently in a previous session a servicelist was created and from now on it should be used as the default one (The original default one will just remain to exist);
3. *Create new servicelist*, i.e. an empty file will be created that is used during this program invocation as "the" servicelist.
4. *Clear the default servicelist*, with the obvious meaning;
5. *Forget it* just in case you have econd thoughts, in the coming session the default list is used.

The database In between program invocations, the data of the services list is maintined in a file. As mentioned in the previous paragraph, more "database"s may exist. The directory where these lists are kept is the directory *serviceLists*, a subdirectory in the directory where Qt-DAB stores its data.

The format in which the data is stored is xml, and the services are stored per channel.

```
<serviceList>
<channel channelName="5B" countryName="NL" ensembleName="5B Z-H/Zeeland ">
  <serviceDesc SID="841b" isFavorite="0" serviceName="Omroep Zeeland " subChId="1"/>
  ...
  <serviceDesc SID="841a" isFavorite="0" serviceName="Radio Rijnmond " subChId="4"/>
</channel>
<channel channelName="9C" countryName="NL" ensembleName="9C ">
  <serviceDesc SID="8816" isFavorite="0" serviceName="Radio Maria " subChId="7"/>
  ...
  <serviceDesc SID="88c7" isFavorite="0" serviceName="538 GreatestHits" subChId="4"/>
</channel>
<channel channelName="12C" countryName="NL" ensembleName="NPO ">
  <serviceDesc SID="8202" isFavorite="1" serviceName="NPO Radio 2 " subChId="1"/>
  <serviceDesc SID="8204" isFavorite="1" serviceName="NPO Klassiek " subChId="3"/>
  <serviceDesc SID="8212" isFavorite="0" serviceName="NPO SterrenNL " subChId="8"/>
  ...
</channel>
...
</channel>
</serviceList>
```

4 Disclaimer

Qt-DAB is developed as hobby program in spare time. Being retired I do have (some) spare time and programming Qt-DAB (and my other programs) is just hobby. It is important to notice that Qt-DAB is - both the sources as the precompiled versions - available *as is*. While it is - obviously - nice if the Qt-DAB software is useful, there are no guarantees, however. If the software does not fit your purpose, or you have problems building an executable, recall that

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